

A Case Study of Customer Communication in Globally Distributed Software Product Development

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ABSTRACT

Agile methodologies emphasize the importance of customer and highlight that communication with the customer is seen as a central tenet. Despite the prevalence and benefits associated with globally distributed software development, communication can be much more challenging in such environments due to temporal, geographical and cultural distances. This paper describes results from a case study of customer communication in a globally distributed product development program applying both traditional and agile approaches. Based on the study, the frequency of communication, the content of that communication and engagement with the customer along with a support for rapid decision making among the development teams and the customer groups are important success factors in distributed agile development.

Categories and Subject Descriptors

D 2.9 [Software Engineering]: Management – *software process models*.

General Terms

Management, Measurement, Documentation, Performance, Design, Human Factors

Keywords

Distributed software development, agile software development, customer communication.

1. INTRODUCTION

Effective communication is often considered as one of the most important contributors to the success of software development [1, 2]. Communication is further emphasized in Agile Software Development (ASD), and is one of the primary motivations underpinning many agile practices such as co-location, stand up meetings and on-site customers [1, 3]. Globally distributed

software development (GSD) is establishing itself as a commonplace approach in the industry e.g. [4, 5] as a trend driven by a number of potential benefits explained e.g. in [5, 6]. In this study, the customer is defined as a “*the source of information relevant for software development team(s) in order to successfully implement the software product*”. Since effective customer communication and continuous feedback are crucial in agile software development (e.g. [1]), the challenges related to customer communication in a globally distributed agile program can be considered extremely important and so several solutions have been proposed, e.g. [7, 8]. In our previous work we discussed the challenges related to the distributed customer communication between traditional and agile development organizations [9, 10]. These challenges were related to restrictive organizational environment resulting in deliberate information hiding by the traditional customer organization, inadequate involvement and support of the customer from the traditional organization, and to the differences between traditional and agile development approaches [9, 10]. However, the previous study focused on only one customer communication aspect of the large ASD program. In this study from the same context reported in [9, 10] the goal is to answer the research questions on *how the customer communication vary between the previously analyzed Irish – US organizations and the other Irish- India and collocated Irish development units and why customer communication was more successful within these units than in the environment reported in [9]*. In order to answer to the research questions, we conducted a case study in which the data was collected in several semi-structured interviews, workshops and regular meetings during the years 2007-2009. This paper is organized as follows. Section 2 introduces the background of the study. The research context is presented in section 3. Section 4 presents the results of the study.

2. BACKGROUND

Software development can be characterized as collaborative, cooperative and complex work that involves a multitude of skills, knowledge and roles. Therefore, efficient communication plays a significant role in any software development practice and it can be considered as one of the most important factors in determining software development success [1, 2]. ASD encourages effective informal communication, fostering physical proximity which is essential for informal communication [1, 11]. One of the characteristics that often distinguish traditional and agile methods is the more detached role of the customer. While agile

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relies on informal active face to face communication with the customer, traditional methods approach customer communication from “as needed” basis [12, 13]. According to Misra et al. [14] successful agile teams should be let of their own allowing them to make their own decisions. The customer needs to be involved in the meetings in order to enable fast decision making (for example about the product backlog items and user stories). However, an agile project team may not always have the luxury of constant or even active customer participation for several reasons, e.g. [15, 16].

Corporate culture is also considered as a success factor for agile development [14]. A successful agile organization should have a culture that supports rapid communication, trust between people, and obtaining quick feedback from the customers [14]. Therefore, a bureaucratic slowly moving organization can be identified as a challenging environment for ASD [17]. One of the most significant barriers to the success seems to lie in changing the whole organization, as opposed to the particular development team in question [12, 18]. This change should be made without compromising both corporate and agile process culture, as stated by Grossman et al. [18].

3. RESEARCH DESIGN

A longitudinal case study was conducted [19] in a large GSD organization. A case study approach was selected because it is an appropriate in situations when actors of the case are important and the context of the case organization is critical [20]. We analysed the use of different communication media using Media Richness Theory (MRT) [21, 22], which proposes that different media are more efficient conveying different information and the selection of the medium should be aligned with the needs of the task. In addition, we used the categorization of communication media by Cockburn [3] to supplement MRT. The analysis framework was similar to the framework presented in [9]. The unit of analysis in this study consists of three different units of the case organization: two Irish collocated development teams (Unit 1), one development team and customer organization in the U.S. (Unit 2) and the development team in Ireland and the quality engineering group in India (Unit 3). Unit 2 has been previously analysed in [9, 10]. The project organization and customer communication environment including the introduced units of the case project is presented in figure 1.

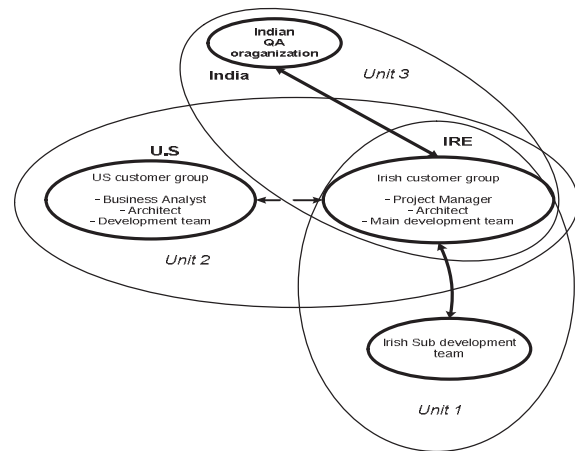


Figure 1. The project organization and customer communication environment of the case project.

The analyzed software development program made both front-end and back-end systems. The front-end was implemented in the U.S. while the Irish development groups were responsible for the implementation of the back-end. The Unit 1 consisted of two separate collocated development teams (main development team and sub development team). The five developers in the main development team were responsible for implementing the important parts of the back-end while the sub-team, consisting of four junior developers implemented simpler tasks. The Irish architect together with the project manager and developers from the Irish main development team formed an onsite customer group for the Irish sub development team. The U.S. organization in Unit 2 including the business analyst and user group formed the customer group for the Irish Main development team, providing them with software requirements. The U.S. customer organization followed traditional waterfall development approach, while agility was adopted only by the Irish and Indian sites (Unit 1 and Unit 3). In Unit 3, two testers from India were working continuously together with the representatives of the Irish main team, architect and project managers.

The use of agile practices within the Irish teams included all Scrum practices and some selected XP practices such as 40 hour week, simple design, collective code ownership and refactoring were used. More detailed description of the applied practices can be found from [10]. Agile practices were not used in the interface between the U.S. customer and Irish main development team. The Indian QA organization was using an agile approach, but the actual practices are unknown due to the lack of detailed information from this context. However, the Indian organization was integrated to the Irish development organization through agile planning, release and daily meetings. The main sources for the analysis were five recorded interviews made with a project manager, architect, and seven developers from Irish main and sub development teams. These interviews followed semi-structure approach with a set of questions which aimed to identify the communication media used in customer communication, what topics were discussed using those media and how often the media was used. These questions were derived from existing literature. Furthermore, the interview framework

included questions related to the challenges encountered in customer communication. The interviews were transcribed and then analysed against the propositions of MRT. Additional source for the analysis were previously made four interviews and observations that were regularly made in seven meetings and six workshops with the case company representatives during the years 2007-2009.

4. FINDINGS

This section analyses two aspects: how the customer communication was conducted between the different units in the case context, and what were the reasons why customer communication was according to the developers observed to be more successful within Unit 1 and Unit 3 than in the Unit 2 reported in [9]. In addition, practical guidelines are presented based on the findings.

Face to face communication occurred only between the two collocated Irish teams (Unit 1). The sub development team had an on-site customer who also was bringing prioritized list of the requirements to the sprint planning meetings. *“We have one business sponsor who basically funds all the work that we do. He is providing facilities to a number of different business (-) and they may have different requirements and different priorities, so we’d work in a group consensus as to what we would do next.”* (Developer). After each sprint, the team also presented the results to the on-site customer in order to get feedback about the work that they have done *“once we have something available we also do a demo for the users [on-site customer] so that we know that we are on the right track”*. (Developer). This finding indicates that the customer was involved in the development process and was providing necessary feedback for the teams. It was relatively easy for the developers of the Irish sub development team to access relevant information when ever needed: *“What was easy about the communication is I can just walk down the corridor and speak to somebody”* (Developer). Apparently, there was no information hiding described in [9] present in the context of Unit 1. In addition to face to face communication, also **wiki** and **email** were actively used.

Our observations support the argument that the agile practices are the most successful in the environments where **rapid communication** is enabled. The idea is that rapid communication is likely to cut down the amount of time spent on major decisions. In the Unit 2 the U.S. customer group was involved in the decision making only in the beginning of the project when the goal was to define fixed up-front requirements for the overall product. After the first round of requirements definition, the customer group did not participate to regular iteration planning activities or daily meetings. This led into the situation in which the requirements had to be refined by the Irish main team in isolation based upon what they believed that the customer needed.

However, the **regular agile meetings** (Sprint planning, release, and daily meetings) were held within Unit 1 and between Unit 1 and Unit 3: *“So it’s a phone call meeting, one person in India calling in as well”*. In those frequent meetings, the purpose was to decide on the goal and contents of the iteration. Altogether, the customer groups from Units 1 and 3 were involved in most of the agile meetings helping the development teams to make rapid

decisions about the goals and user stories that would be developed during the next iteration. All these meetings between Ireland and India were held through **telephone**. In addition, **wiki** and **email** were actively used for exchanging information within the Unit 3. Naturally, the practice of having an on-site customer can be seen as a key element in active customer participation. However, the Irish architect worked as a customer also for the Indian QA unit (Unit 3). We did not found out major challenges, such as information hiding and lack of customer involvement, in the communication between the Indian QA organization and the Irish main team. In this case agile meetings increased the transparency of the work and enabled information sharing between India and Ireland. Agile meetings can be seen as a means to increase transparency and visibility between the stakeholders, so the involvement of customer in the context of Unit 3 was significantly better than in the case of Unit 2 where it was virtually non-existent [9]. **Videoconferencing** was only occasionally used. Figure 2 presents the customer communication channels and their usage in the case project. The dotted line between U.S. and IRE indicates that this communication relationship has already been analyzed in [9, 10].

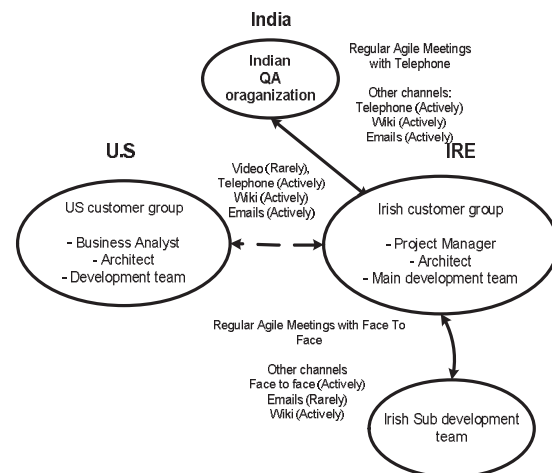


Figure 2. The communication channels and their usage in the case project.

5. CONCLUSIONS

The answer to the first research question is that there were no major differences between the activeness of customer communication in the context of Unit 1 and Unit 3 when compared to Unit 2. In all three cases the customer communication was active and similar communication channels varying in their capability of conveying different kinds of information were used. The only difference in communication media between the *distributed* units was the use of videoconferencing within Unit 2 that was lacking from Unit 3. The collocated Irish teams were able to utilize face to face communication. The most interesting comparison is thus between the units 2 and 3. Unit 3 was following an agile approach and the members of Unit 3 were integrated to the development process through regular planning, release and daily meetings, despite the time-zone differences. In addition, there were no indications of information hiding and the lack of customer involvement in the

relationship between the Indian QA organization and Irish development organization.

Compared to Units 1 and 3, the culture of the U.S. customer group in Unit 2 can be considered bureaucratic and also slower in reacting partially due to the transition from traditional to agile approach. [10]. The main challenges realized in form of an uninvolved customer and information hiding which significantly hindered customer communication in the context of Unit 2. Thus, **rapid communication, regular agile meetings and involved customer** can be seen as factors contributing to the more successful communication between the Unit 1 and Unit 3 when compared to the Unit 2. These findings answer the second research question of this paper.

Based on our case study we revealed some customer communication related lessons that the industry could take into consideration when adopting an agile approach in the globally distributed context. For instance it seems that 1) the agile development process should be taken into account at the program level approach instead of using it only between some organizational units, 2) customer responsibilities in the context of applying both traditional and agile development approaches should be more carefully defined, the company should assure that 3) the roles in the defined customer group are clear and that 4) at least some representatives from the customer group is always involved in the agile meetings. This study has its limitations. We were not able to interview stakeholders from either the U.S. or Indian organizations. Thus, the results are drawn from the perspective of the Irish stakeholders alone which creates a narrower and more limited view to the phenomenon observed. In addition, due to the confidentiality reasons access to more precise data such as email conversations and project wiki was not given. Even though all the possible information sources could not be accessed, as many of them as possible under the given circumstances were utilized and the obtained data was analysed by all the participating researchers. Thus, *data triangulation* and *investigator triangulation* as introduced by Stake [23] were used in order to increase the validity of the results.

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